

Session 5 · Deck 5.B3

Chebyshev, and Smolyak as a Basis

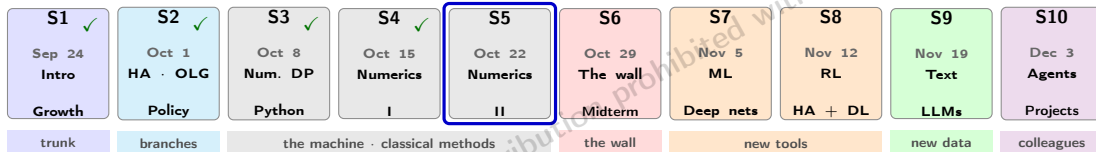
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Where We Are

you are here: 5.B3 ▼



Build it on paper → solve it on the machine → push it to the frontier with AI

Block B: 5.B1 the residual · 5.B2 choosing a basis · **5.B3** Chebyshev · 5.B4 finite elements · 5.B5 determining the coefficients · 5.B6 the worked example.

One Family, Two Famous Members

The **Jacobi** (or hypergeometric) family $P_n^{a,b}$, for $a, b > -1$, is defined by an orthogonality condition with a weight:

$$\int_{-1}^1 (1-x)^a (1+x)^b P_n^{a,b}(x) P_m^{a,b}(x) dx = 0, \quad m \neq n.$$

The weight decides which family you get, and $a, b > -1$ keeps the integral finite.

- $a = b = -\frac{1}{2}$: **Chebyshev**. The weight blows up at the endpoints, so the family pays extra attention there.
- $a = b = 0$: **Legendre**. Uniform weight; the right family for Gauss–Legendre quadrature, which 4.B2 already used.

Quant_Macro Lec. 9, "Orthogonal Polynomials".

Notation

The literature writes the Jacobi parameters as α, β . In this course α is the capital share and β the discount factor, so here they are a, b .

That endpoint weighting is not a curiosity. It is why Chebyshev nodes cluster at the ends, and why they defeat Runge's phenomenon (4.B1).

Why Chebyshev Is the Default

- Many **simple closed-form expressions** are available.
- More **robust than the alternatives** for interpolation.
- **Bounded** in $[-1, 1]$ — so no coefficient has to cancel a large number, which is precisely what killed the monomials.
- **Smooth.**
- Moving between coefficients and values at the Chebyshev nodes is a **cosine transform** — an FFT, $O(n \log n)$.
- Several **theorems bound the interpolation error**, so the choice can be defended rather than asserted.

Standard references: Boyd (2001), *Chebyshev and Fourier Spectral Methods*; Fornberg (1998), *A Practical Guide to Pseudospectral Methods*.

Boyd's own rule of thumb is worth carrying: **when in doubt, use Chebyshev** — unless the solution is periodic, in which case use Fourier.

The Recursion and the Nodes

$$T_0(x) = 1$$

$$T_1(x) = x$$

$$T_{n+1}(x) = 2x T_n(x) - T_{n-1}(x)$$

giving $1, x, 2x^2 - 1, 4x^3 - 3x, 8x^4 - 8x^2 + 1, \dots$

The n roots of T_n are

$$x_k = \cos \frac{2k-1}{2n} \pi, \quad k = 1, \dots, n.$$

They are **not** equally spaced: they cluster quadratically towards ± 1 , which is exactly where an equispaced interpolant misbehaves.

Quant_Macro Lec. 9, "Chebyshev polynomials" (recursion frame).

Where you have seen this

4.B1 measured it: on Runge's function, equispaced interpolation *diverges* as the degree rises, while Chebyshev nodes converge. Same nodes, and now the polynomials that go with them.

Three lines of code. The recursion is numerically stable, which is more than can be said for computing x^n directly.

Closed Forms, and the Change of Variable

$$T_n(x) = \begin{cases} \cos(n \arccos x) & -1 \leq x \leq 1 \\ \cosh(n \operatorname{arcosh} x) & x \geq 1 \\ (-1)^n \cosh(n \operatorname{arcosh}(-x)) & x \leq -1 \end{cases}$$

A polynomial written as a cosine: that identity is the reason the coefficient transform is an FFT, and the reason $|T_n| \leq 1$ on the interval.

Our state space is not $[-1, 1]$, so map it in:

$$x \mapsto 2 \frac{x-a}{b-a} - 1.$$

The weight

Chebyshev polynomials are orthogonal with respect to

$$w(x) = \frac{1}{\sqrt{1-x^2}},$$

which is infinite at ± 1 — the analytical statement of the same endpoint emphasis.

Choosing $[a, b]$ is a modelling decision, not a formality: outside it the polynomial is not just inaccurate, it diverges like cosh.

The Interpolation Theorem

Chebyshev Interpolation Theorem

If an approximating polynomial is made exact at the roots of T_n , then as $n \rightarrow \infty$ the approximation error becomes arbitrarily small — for any continuous function.

So you are within a factor of three of **the best any polynomial of that degree could do** — and the best one is not computable, while this one is an FFT.

There is a sharper version worth knowing, because it is what makes the choice *defensible* rather than merely traditional. Chebyshev-node interpolation is never much worse than the **best possible** polynomial of the same degree:

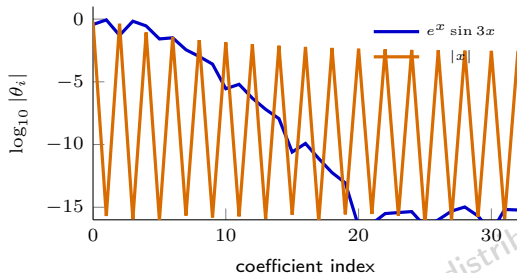
$$\|f - p_n\|_\infty \leq \left(\frac{2}{\pi} \log(n+1) + 1 \right) \|f - p_n^{\text{best}}\|_\infty.$$

At $n = 33$ that factor is **3.24**.

The factor grows like $\log n$, so it is still 4.0 at $n = 1000$. It never bites.

Quant_Macro Lec. 9, "Chebyshev polynomials" (interpolation-theorem frame). Ledger chebyshev.lebesgue_factor_n33 = 3.2450.

Measured: What Smoothness Buys, and What a Kink Costs



Each line is one function's list of Chebyshev coefficients θ_i : $e^x \sin 3x$ is analytic, $|x|$ has a corner mid-interval. Truncating at n drops the tail, so the error is about the first coefficient dropped — the picture predicts the table.

The sawtooth is not noise. $|x|$ is even and the odd T_i are odd, so every odd coefficient is exactly zero — the orange line alternates between even coefficients and zeros.

n	$e^x \sin 3x$	$ x $
9	$10^{-3.6}$	$10^{-1.2}$
17	$10^{-11.1}$	$10^{-1.5}$
33	$10^{-14.3}$	$10^{-1.7}$
65	$10^{-13.8}$	$10^{-2.0}$

max interpolation error

Ledger chebyshev. The analytic case stops improving at $n = 65$ because it has reached 10^{-14} — rounding, not approximation.

More Than One Dimension: The Tensor Product

Chebyshev polynomials live on an interval; economic state spaces do not. The natural generalization multiplies them out. To approximate $F : [-1, 1]^d \rightarrow \mathbb{R}$ at degree κ ,

$$\hat{F}(x) = \sum_{n_1=0}^{\kappa} \cdots \sum_{n_d=0}^{\kappa} \theta_{n_1 \dots n_d} T_{n_1}(x_1) \cdots T_{n_d}(x_d).$$

The good news. If the one-dimensional basis is orthogonal under a weight, the tensor-product basis is orthogonal under the product weight. Everything from the one-dimensional theory survives.

The bad news is the number of sums.

Quant_Macro Lec_9: "Tensors and Chebyshev Polynomial Approximation" and "Curse of Dimensionality in Tensor Approximation".

Count them

Each index runs over $\kappa + 1$ values, and there are d of them:

$(\kappa + 1)^d$ coefficients.

This is the same curse 3.A2 met with grids, arriving through a different door. Changing the representation did not remove it.

The First Fix: Drop the High-Order Interactions

The tensor basis includes $x_1^\kappa x_2^\kappa \cdots x_d^\kappa$ — a term of total degree $d\kappa$. Nobody needs that.

Complete polynomials keep only the terms whose exponents sum to at most κ :

$$P_\kappa^d \equiv \left\{ x_1^{i_1} \cdots x_d^{i_d} \mid \sum_{j=1}^d i_j \leq \kappa \right\}.$$

Instead of letting every variable reach degree κ independently, cap the **total** degree. Only low-complexity interactions survive — which are the ones that matter for a smooth function.

The count falls from $(\kappa + 1)^d$ to $\binom{\kappa + d}{d}$.

d	tensor	complete
2	25	15
3	125	35
5	3,125	126
10	9,765,625	1,001

$\kappa = 4$

Still exponential?

No — $\binom{\kappa + d}{d}$ is polynomial in d for fixed κ . But it is still large, and it grows fast in κ .

Quant_Macro Lec. 9, "Complete Homogeneous Symmetric Polynomials". Gaspar & Judd (1997), "Solving Large-Scale Rational-Expectations Models,"

Macroeconomic Dynamics 1(1), 45–75. Ledger chebyshev.tensor_vs_complete.

Which Terms Survive: $d = 2, \kappa = 2$

Two variables, degree two. Write each term as $x_1^{i_1} x_2^{i_2}$ and sort by **total degree** $i_1 + i_2$:

$i_1 + i_2$	term	kept?
0	1	both
1	x_1, x_2	both
2	$x_1^2, x_1 x_2, x_2^2$	both
3	$x_1^2 x_2, x_1 x_2^2$	tensor only
4	$x_1^2 x_2^2$	tensor only

The tensor rule caps each index separately and keeps all $3 \times 3 = 9$. The complete rule caps the *sum* and keeps the first three rows — **6**.

Ledger chebyshev.tensor_vs_complete: $9 \rightarrow 6$ at $d = 2$ and $27 \rightarrow 10$ at $d = 3$, both with $\kappa = 2$.

The rule in one line

Keep $x_1^{i_1} \cdots x_d^{i_d}$ if $\sum_j i_j \leq \kappa$. Throw away everything else.

Same rule at $d = 3, \kappa = 2$: the tensor basis keeps $3^3 = 27$ terms, the complete basis $\binom{2+3}{3} = 10$. The gap widens with every dimension added.

What you give up are the **high-order interactions** — for a smooth function, the smallest terms anyway, which is why the trade is usually free.

Smolyak Returns — and It Is a Different Object Now

4.B1 built Smolyak sparse grids as a **node set**: where to evaluate a function you are interpolating, and where to put quadrature nodes. The construction was the nested ladder $m_1 = 1$, $m_i = 2^{i-1} + 1$ of Chebyshev extrema, with $G^1 \subset G^2 \subset \dots$, combined over the index sets $\sum_j i_j \leq d + \mu$.

Here the *same* index rule selects which **polynomials** enter the approximation:

$$\hat{F}(x) = \sum_{\mathbf{i} \in \Phi(q,d)} \theta_{\mathbf{i}} T_{i_1}(x_1) \cdots T_{i_d}(x_d),$$

and the coefficients are chosen to kill a residual rather than to interpolate known values.

Construction, nestedness and the index sets are 4.B1's — see that deck rather than re-deriving them. Smolyak (1963); in economics Krueger & Kubler (2004), Malin, Krueger & Kubler (2011), Judd, Maliar, Maliar & Valero (2014).

Node set $\rightarrow$ basis

Same combinatorics, different job. There it answered *where do I evaluate?* Here it answers *which terms do I keep?* And the signed binomial weights of 4.B1 are what make the two views consistent.

So a Smolyak collocation method uses the sparse set *twice*: as the basis, and as the collocation points.

What It Buys, and What It Costs

4.B1's measured point counts, reused rather than recomputed:

d	Smolyak $\mu = 2$	tensor, 5/dim
2	13	25
5	61	3,125
10	221	9,765,625

At $d = 10$: **221 terms against nearly ten million.**
And the run reproduces Malin, Krueger & Kubler's published table exactly at $\mu = 2$ — 13, 41, 61, 313 for $d = 2, 4, 5, 12$.

The curse is weakened, not broken

For $f \in C^k$ the error bound is

$$C_d n^{-k} (\log n)^{(d-1)(k+1)},$$

the tensor rate spoiled by a logarithmic factor whose exponent **grows in d** . Sparsity postpones the curse; it does not repeal it.

Which is the honest setup for S7: when d is large enough that even this fails, the answer is a basis that *adapts* rather than one chosen in advance.

Ledger smolyak in tools/figures/s04_numbers.json — 4.B1's run, not recomputed here. Error bound: Barthelmann, Novak & Ritter (1999).

Takeaways

- Chebyshev and Legendre are two members of the **Jacobi family**, distinguished by the orthogonality weight. Chebyshev's weight $1/\sqrt{1-x^2}$ emphasises the endpoints — which is why its nodes cluster there.
- Bounded, smooth, stably generated by a three-term recursion, transformed by an FFT, and backed by error theorems. **When in doubt, use Chebyshev.**
- The interpolation theorem, in its sharp form: Chebyshev-node interpolation is within $\frac{2}{\pi} \log(n+1) + 1$ of the *best possible* polynomial of that degree — a factor of **3.24** at $n = 33$.
- Measured: for an analytic function the error reaches **machine precision by degree 33**; for $|x|$ it falls like $1/n$ and sixty-five terms buy two digits. **The basis is not the problem — the function is.**
- The tensor product preserves orthogonality and costs $(\kappa+1)^d$ coefficients. **Complete polynomials** cut that to $\binom{\kappa+d}{d}$: 1,001 instead of 9.8 million at $d = 10$, $\kappa = 4$.
- **Smolyak**, 4.B1's node set, is here a **basis**: the same index rule chooses which polynomials enter. 221 terms at $d = 10$ against nearly ten million — and an error bound that weakens the curse without breaking it.

Bridge: When the Function Has a Corner

Everything in this deck rewards **smoothness**, and the measured penalty for its absence was severe: $|x|$ needed sixty-five Chebyshev terms to reach two digits, and 5.B2 showed the oscillation beside a kink never goes away.

Economics supplies corners as a matter of course — borrowing limits, irreversibility, non-negativity. So there has to be a basis that does not care.

Next (5.B4)

Finite elements: a **local** basis. Chop the domain into pieces, put a simple function on each, and let a piece boundary sit exactly where the kink is. Slow to converge, indifferent to smoothness, and sparse.